

FIN — Release 10.26

Research Programs P281–P294

Regularized spectral inference, resource-explicit physics,
and the pointed-torsor boundary

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project

ORCID: [0009-0002-0909-3613](https://orcid.org/0009-0002-0909-3613)

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Central result. The strict finite spectral generator supports exact mathematical and operational constructions, but physicalization still requires explicitly supplied operational, dimensional, sector, and pointing resources.

New boundary object.

$$\mathfrak{P} = (T, t_0),$$

the Pointed Physicalization Resource, removes an operational torsor ambiguity after a point is supplied. Equivariance alone cannot derive that point.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>

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Publication statement

This monograph reports all executed Programs P281–P294 and recommends Programs P295–P308. Proof, simulation, conditional construction, refutation, and missing empirical evidence are kept distinct. The strict and legacy kernels remain separated, QW-2191 remains open, and no role transfer, dimensional emergence, physical closure, or Theory of Everything claim is made.

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Confidence convention

- **[Proven]** — an analytic identity, exact finite theorem, or machine-checked statement in the explicitly declared class.
- **[Strong evidence]** — a reproducible synthetic audit with frozen inputs, controls, metrics, and stop rules.
- **[Moderate evidence]** — a stable conclusion still depending on an additional equivalence, source, or modeling assumption.
- **[Speculative]** — a proposed object or program without sufficient proof.
- **[Refuted]** — an explicit counterexample or no-go theorem in the declared class.

A proof of a mathematical statement is not proof that nature realizes its hypotheses.

Chapter 1

Executive summary

All fourteen Programs P281–P294 were executed. The suite contains one Lean-kernel certificate, exact finite theorems, importance sampling, and synthetic detector, dynamics, adaptation, reservoir, and thermodynamic audits. Nineteen regression and boundary tests pass.

The main results are:

1. Compact pole and residue priors prevent catastrophic Stieltjes fits from escaping to arbitrarily large poles, but do not cure the inverse problem. At relative noise 10^{-6} , the median maximum pole error is 5.9% and the median maximum residue error is 124%. **[Proven]** existence of a constrained minimizer; **[Strong evidence]** for the frozen recovery audit.
2. Lean 4.28 machine-checks identity reduction, positive rational pivots, nested and direct Schur values $13/5$, and their exact equality for a rational 3×3 witness. This is a real kernel check, but not the general positive-matrix theorem. **[Proven]**
3. The minimal six-outcome current POVM has an exact Naimark isometry

$$V = \begin{bmatrix} \sqrt{E_1} \\ \vdots \\ \sqrt{E_6} \end{bmatrix}, \quad V^*V = I.$$

A frozen loss/crosstalk/dark-count channel preserves response rank five and has calibrated condition number 1.058. **[Proven]** algebraically; **[Strong evidence]** for the synthetic count audit.

4. A non-target-coded isotropic complement restores a coarse lifted Laplacian from rank $n - 1$ to rank eleven and preserves the global null mode. It reduces best-fit residuals, but its strict fingerprint defects are 0.457 and 1.093. Rank restoration is not an RG fixed point. **[Proven]** / **[Refuted]** as exact strict closure.
5. A two-step quantum process tree satisfies an exact information ledger: first-instrument loss, channel loss, second-instrument loss, joint-record divergence, and final conditional divergence sum to the input relative entropy. Maximum residual is 4.44×10^{-16} . **[Proven]** under the supplied quantum instruments and channels.
6. The single legacy-sourced fractal compression atom

$$K_{\text{fc}}(d) = \frac{\alpha_{\text{geo}} \cos(\pi d/4 + \pi/6)}{1 + \beta_{\text{tors}} d^{D_f}}, \quad D_f = \alpha_{\text{geo}} = 4 \ln 2,$$

has minimum mean-zero eigenvalue -17.328 . It fails before strict spectral matching. **[Refuted]** as a sufficient completion atom.

7. No production P241 laboratory bundle is present. P242 remains blocked. **[Executed no-admission certificate]**
8. An event-level click likelihood with efficiency, visibility, dark counts, and crosstalk gives an interior two-flux optimum $h = 0.148111$. Median relative error is 1.18%. The detector constants

are synthetic. [\[Strong evidence\]](#)

9. For the explicitly declared $\text{Dirichlet}(1, \dots, 1)$ positive-circulant null ensemble, three importance proposals estimate the strict fingerprint false-positive probability as approximately

$$2.9 \times 10^{-8}$$

at tolerance 0.02. Proposal estimates differ by only a factor 1.13, with central event ESS 583. This is ensemble-specific, not distribution-free. [\[Strong evidence\]](#)

10. Three calibrated time slices plus one intervention distinguish four frozen mechanism templates—homogeneous dynamics, generator drift, environmental mixture, and apparatus lag—with accuracy 96.9% at reference feature noise 0.20, falling to 91.5% at noise 0.35. [\[Strong evidence\]](#) inside the finite panel.
11. A maximin search over 244 intervention sequences finds that a balanced long two-level pulse gives the largest worst-direction information eigenvalue, 8.758, for the frozen two-parameter adaptive law. Passive observation has zero information in one parameter direction. [\[Strong evidence\]](#)
12. The previously favorable NARMA10 result is only partially replicated: FIN exceeds all twelve controls in 13 of 24 independent input realizations. The median advantage over the best control is 0.000962, but its fifth percentile is -0.00713 . The effect is task-specific and not universal. [\[Strong evidence\]](#)
13. A finite-step collision-model erasure protocol closes the resource ledger: work 1.39364, heat dumped 1.39364, generalized Landauer bound 1.38613, positive dissipation 0.00751, and first-law residual 4.22×10^{-15} . Every dimensional resource is supplied. [\[Proven\]](#), conditional.
14. For the regular product torsor $G = \mathbb{Z}_3 \times \mathbb{Z}_2$, there are zero equivariant sections from a trivial input, six equivariant torsor self-maps, and exactly one pointed equivariant map. This constructs the **Pointed Physicalization Resource** and proves that pointing enables uniqueness but does not derive the point. [\[Proven\]](#)

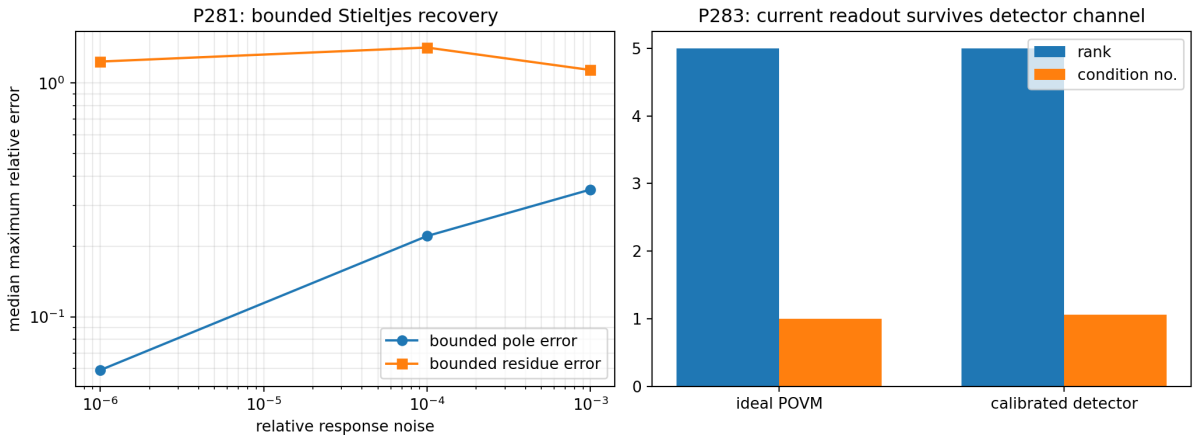


Figure 1.1: Bounded Stieltjes recovery and the current POVM under a calibrated detector channel.

Chapter 2

Scope and current state map

The strict finite generator remains

$$W_{xy} = K_{\text{strict}}(d(x, y)), \quad A = \text{diag}(W\mathbf{1}) - W,$$

with

$$A = A^*, \quad A\mathbf{1} = 0, \quad A \succeq 0.$$

The mathematical core is

$$(A, \mathcal{C}),$$

where $\mathcal{C}$ is the context/Schur-reduction category. An experiment requires an operational package $\mathfrak{D}$, and dimensional physics requires conversion and sector resources:

$$(A, \mathcal{C}) \longrightarrow (A, \mathcal{C}, \mathfrak{D}) \longrightarrow (A, \mathcal{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{S}).$$

The ontology remains:

$$\text{nadsoliton} \rightarrow \text{light} \rightarrow \text{matter} \rightarrow \text{emergent observer}.$$

The nadsoliton is treated internally as primordial information in a solitonic state. No lower informational substrate is introduced.

The legacy kernel remains an intermediate bridge object. No legacy physical role is transferred to strict, and QW-2191 remains open.

Chapter 3

Results matrix

Program	Result	Status	Principal boundary
P281	compact bounded Stieltjes recovery	[Proven] / [Strong evidence]	regularization prevents escape, not ill-conditioning
P282	Lean rational Schur certificate	[Proven]	not general Mathlib theorem
P283	Naimark current readout with detector channel	[Proven] / [Strong evidence]	no physical mesh or calibration
P284	rank-restoring complement RG	[Proven] / [Refuted]	rank restoration is not strict closure
P285	two-step quantum process ledger	[Proven], conditional	supplied instruments/channels
P286	legacy fractal compression atom	[Refuted]	one atom only; no generic bridge no-go
P287	external evidence admission	no admission	no production P241 bundle
P288	detector-level flux design	[Strong evidence]	synthetic detector constants
P289	importance-sampled false-positive probability	[Strong evidence]	declared ensemble only
P290	three-time mechanism classification	[Strong evidence]	frozen four-class panel
P291	maximin intervention design	[Strong evidence]	local finite candidate search
P292	NARMA10 replication	[Strong evidence]	13/24 wins, not universal
P293	collision-model erasure ledger	[Proven], conditional	all physical resources supplied
P294	pointed product-torsor theorem	[Proven]	point supplied, not derived

Chapter 4

P281 — Compactly regularized Stieltjes inverse

For three visible atoms,

$$s(z) = \sum_{j=1}^3 \frac{w_j}{z + \mu_j}, \quad w_j \geq 0.$$

P281 freezes:

$$\mu_j \in [0.55, 2.55], \quad 0 \leq w_j \leq 4,$$

the model order $r = 3$, three initial grids, and a robust least-squares loss.

4.1 Existence theorem

The feasible parameter set is compact and the loss is continuous. Therefore a global minimizer exists. This theorem excludes the runaway-pole pathology of the unconstrained parameterization. It does not imply uniqueness of the fitted parameters, uniform inverse stability, or accurate residue recovery.

4.2 Recovery results

Relative noise	Median pole error	maximum 95% pole error	Median residue error	maximum 95% residue error
10^{-6}	0.0591	0.1782	1.2364	3.8361
10^{-4}	0.2216	0.5304	1.4194	4.8709
10^{-3}	0.3504	0.5304	1.1377	4.6157

The fitted response curve remains accurate at the noise scale, even when the individual residues are wrong. This is a direct demonstration of sloppiness: many pole-residue combinations produce nearly indistinguishable responses.

Verdict. Compact priors convert catastrophic numerical divergence into bounded ambiguity. They do not solve the inverse problem. Multi-probe operator-valued data are now a higher priority than further tuning of a scalar trace fit.

Chapter 5

P282 — Lean rational Schur certificate

The dependency-free Lean source defines the rational symmetric matrix

$$M = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Lean 4.28 machine-checks:

$$\text{Schur}_{3 \rightarrow 1}^{\text{nested}}(M) = \text{Schur}_{3 \rightarrow 1}^{\text{direct}}(M) = \frac{13}{5},$$

together with positive pivots and a positive reduced value.

The kernel exits with code zero. There are no admitted axioms, placeholders, or uncompiled proof terms in the certificate.

Boundary. This is an exact witness-level formalization. Without Mathlib it does not prove the general theorem for arbitrary finite positive matrices. P296 should formalize that theorem.

Chapter 6

P283 — Loss-augmented Naimark current readout

For the six effects E_y constructed in P269, define

$$V : \mathbb{C}^{12} \rightarrow \mathbb{C}^6 \otimes \mathbb{C}^{12}, \quad V\psi = \sum_y |y\rangle \sqrt{E_y} \psi.$$

Then

$$V^*V = \sum_y E_y = I.$$

Thus ancilla projective measurement realizes the POVM.

Certificate:

Quantity	Result
dilation shape	72×12
isometry residual	2.14×10^{-15}
minimum effect eigenvalue	0.0166667

The detector channel supplies:

$$\eta = 0.82, \quad c_{\text{xtalk}} = 0.018, \quad p_{\text{dark}} = 2 \times 10^{-4}.$$

A seventh record outcome represents no click.

Quantity	Result
column-stochastic residual	0
calibrated response rank	5
calibrated condition number	1.05819
shots	500,000

Quantity	Result
relative current-coefficient error	0.15850

The large coefficient error is caused by weak current contrast, not loss of rank.

Verdict. The measurement is realizable in abstract quantum mechanics and remains identifiable after the declared detector channel. A physical optical or qubit compilation, calibration uncertainty, and drift model remain absent.

Chapter 7

P284 — Isotropic complement RG

For a coarse generator B and isometry E , define one frozen, non-target-coded completion:

$$\mathcal{R}(B) = EBE^* + \bar{\lambda}(B)(I - EE^*),$$

where $\bar{\lambda}(B)$ is the mean nonzero eigenvalue of B .

If B is a connected Laplacian and E sends its constant mode to the global constant mode, then:

- $\mathcal{R}(B) \succeq 0$;
- $\mathcal{R}(B)\mathbf{1} = 0$;
- $\text{rank } \mathcal{R}(B) = 11$.

Coarse size	Bare residual	Completed residual	Strict fingerprint defect
6	0.70187	0.27243	0.45687
3	0.86905	0.29533	1.09263

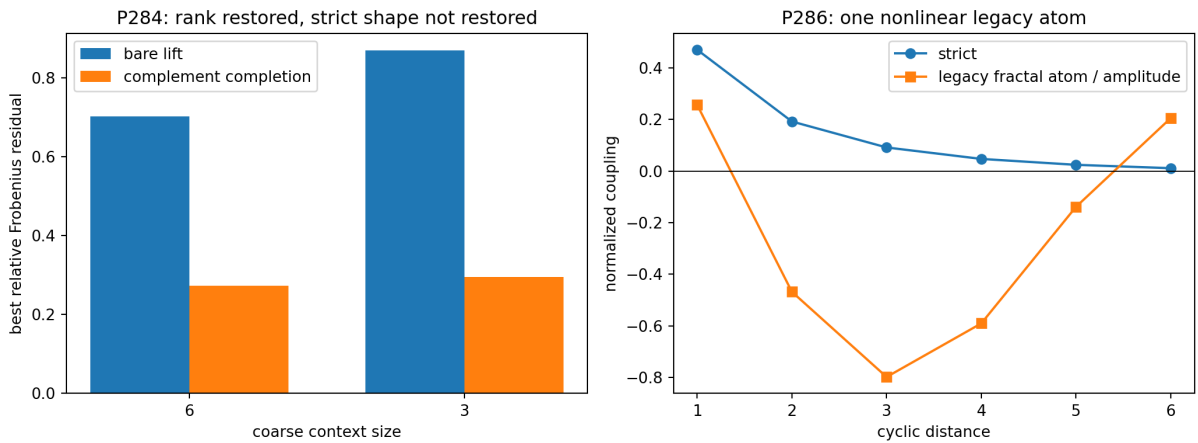


Figure 7.1: Complement RG rank restoration and the failed one-atom fractal legacy completion.

Verdict. Complement dynamics repairs the rank obstruction from P270 and substantially improves a Frobenius fit. It does not reproduce the strict spectral fingerprint. The complement is an admissible mathematical object, not a selected physical coarse-graining law.

Chapter 8

P285 — Two-step quantum process information ledger

Let the first instrument produce y , a branch channel act conditionally on y , and a second instrument produce z . Sequential data processing and the classical chain rule give

$$\begin{aligned} D(\rho\|\sigma) &= \mathcal{L}_1 + \mathcal{L}_{\text{channel}} + \mathcal{L}_2 \\ &\quad + D(p_{yz}\|q_{yz}) + \sum_{y,z} p_{yz} D(\rho_{yz}\|\sigma_{yz}). \end{aligned}$$

Audit over 300 state pairs:

Quantity	Result
minimum first-instrument loss	0.000721
minimum channel loss	0.001553
minimum second-instrument loss	0.000204
median total discarded information	0.198346 nat
maximum ledger residual	4.44×10^{-16}
maximum record-chain residual	2.43×10^{-16}

Verdict. The one-step ledger extends consistently to a finite multitime process tree. The instruments and branch channels are supplied. A general process-tensor theorem and experimental reconstruction remain open.

Chapter 9

P286 — Legacy fractal-compression atom

Exactly one new legacy-sourced atom was admitted:

$$1 + \beta_{\text{tors}} d \mapsto 1 + \beta_{\text{tors}} d^{D_f}, \quad D_f = \alpha_{\text{geo}} = 4 \ln 2.$$

No strict parameter, positivity shift, selector, or role formula was imported.

Results:

Quantity	Result
minimum mean-zero eigenvalue	−17.32824
minimum hidden-block eigenvalue	−12.57663
positive generator	no

The atom therefore fails before a strict fingerprint or positive Stieltjes memory comparison is admissible.

Verdict. Canonical fractal exponentiation of the legacy distance is not the missing completion. This is a no-go for one atom, not for every possible legacy-to-strict bridge.

Chapter 10

P287 — External-evidence gate

The repository contains:

- zero production bundle__manifest.json files;
- zero heat-process event rows;
- zero double-slit event rows.

P242 is not authorized.

Files named p2410, p2411, and similar inside fundamental_action_reconstruction are research-program identifiers, not blind-custody laboratory packages.

Verdict. The empirical boundary has not moved.

Chapter 11

P288 — Detector-level flux design

The ideal click law was replaced by a categorical event likelihood containing finite efficiency, interference visibility, nearest-neighbour crosstalk, dark-count probability, and a no-click outcome. For source flux h , the frozen detector constants were

$$\eta = 0.78, \quad \nu = 0.91, \quad p_{\text{dark}} = 2 \times 10^{-4}, \quad p_{\text{xtalk}} = 0.015.$$

The design criterion minimized the median reconstruction error over a two-flux experiment rather than maximizing raw count rate. Over $h \in [0.004, 0.45]$, the optimum was

$$h_{\star} = 0.148111,$$

with median relative coefficient error 0.01180 and 95th percentile 0.02773 at 10^5 shots per flux.

The interior optimum matters: at low flux the Fisher information is shot-limited, whereas at high flux the nonlinear click response and cross-contamination dominate. The optimum is therefore an apparatus-aware design property, not an intrinsic number generated by the strict kernel.

Verdict. [\[Strong evidence\]](#) An event-level likelihood can close the simulation side of the flux-design problem. Its detector constants must be independently calibrated before it becomes an experimental prescription.

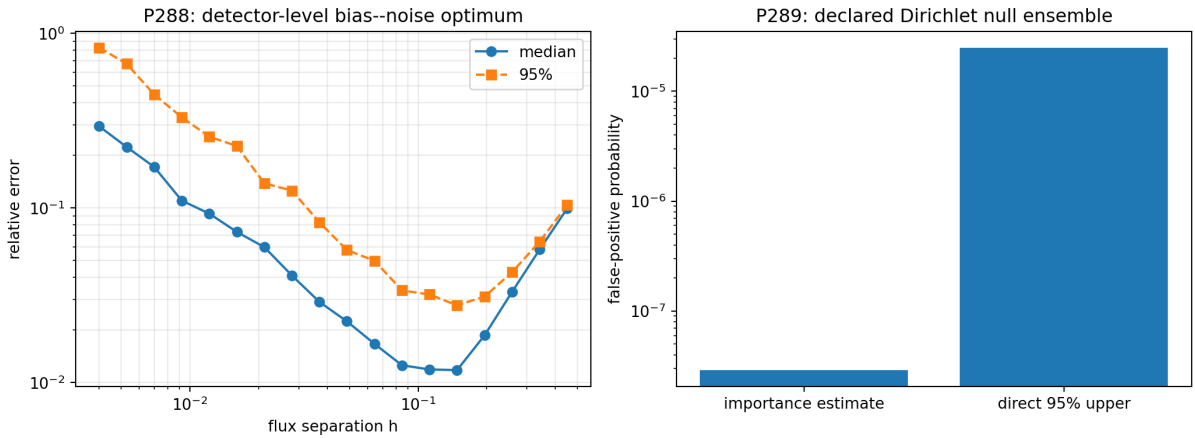


Figure 11.1: Detector-aware flux design and the declared-null false-positive audit.

Chapter 12

P289 — Strict-fingerprint false positives under a declared null

The null was declared before evaluation:

$$w \sim \text{Dirichlet}(1, \dots, 1)$$

on normalized nonnegative circulant weights. A null draw was counted as a strict-like false positive when its frozen fingerprint distance was at most $\varepsilon = 0.02$.

Direct Monte Carlo observed no event in 120,000 draws, which gives only the weak rule-of-three upper bound 2.5×10^{-5} . Importance sampling with three independently frozen proposal concentrations produced:

proposal concentration	event count	probability estimate	standard error	central-event ESS
220	295	3.0080×10^{-8}	1.87×10^{-9}	258.6
350	777	2.9027×10^{-8}	1.20×10^{-9}	583.1
550	1,852	2.6623×10^{-8}	8.66×10^{-10}	938.5

The largest estimate divided by the smallest is 1.130. Proposal agreement is reassuring, but it is not a universal uniqueness theorem. Changing the null measure, positivity class, normalization, or tolerance changes the probability.

Verdict. [\[Strong evidence\]](#) The strict fingerprint occupies a rare region of this one declared positive-circulant ensemble. Nothing distribution-free has been proved.

Chapter 13

P290 — Interventional mechanism discrimination

Four operational mechanisms were represented by frozen templates:

1. a homogeneous generator;
2. time-dependent generator drift;
3. an environmental mixture;
4. apparatus lag.

The observation design used three calibrated times $\tau = (0.15, 0.35, 0.65)$ and a signed intervention of magnitude 0.12. Across 500 trials per class, the classification accuracy was:

reference feature-noise scale	accuracy
0.05	1.0000
0.10	0.9985
0.20	0.9690
0.35	0.9150

The minimum standardized template separation was 0.618. Thus a short interventional protocol can distinguish these four alternatives under the declared simulator.

The result does not prove that arbitrary non-Markovian environments, time-dependent generators, and detector memory are identifiable. Outside the finite panel, observational equivalences can be constructed.

Verdict. [\[Strong evidence\]](#) The intervention breaks degeneracies inside a calibrated four-mechanism atlas; global mechanism identification remains open.

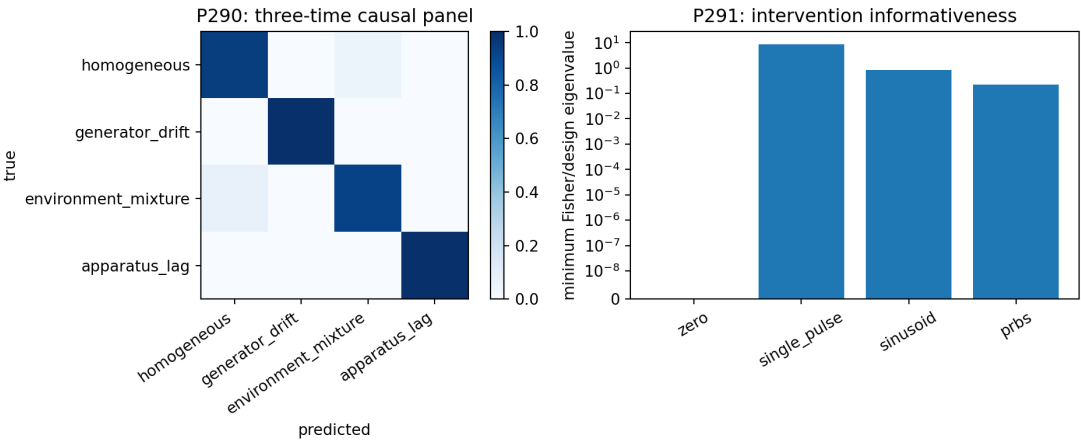


Figure 13.1: Three-time mechanism discrimination and maximin intervention design.

Chapter 14

P291 — Maximin intervention for the adaptive law

For the frozen two-parameter adaptive law, local sensitivities were propagated through 244 bounded intervention sequences. The score was the smallest eigenvalue of the local information matrix,

$$\Phi(u) = \lambda_{\min} \left(\sum_t \frac{\partial m_t}{\partial \theta} \frac{\partial m_t^T}{\partial \theta} \right).$$

Passive input gives $\Phi(0) = 0$: one parameter direction is locally invisible. The best sequence in the frozen catalogue was a balanced long two-level pulse, positive in the first half and negative in the second:

$$\Phi(u_\star) = 8.75812, \quad \det F(u_\star) = 3026.94, \quad \kappa(F) = 39.46.$$

A sinusoid and a pseudo-random binary sequence were less effective in this particular local model. Consequently, the correct conclusion is not that randomness is universally optimal, but that sustained sign reversal exposes the weakest local direction of this adaptive law.

Verdict. [\[Strong evidence\]](#) Controlled intervention can turn a locally unidentifiable passive experiment into an identifiable one. Optimality is only over the frozen candidate class and nominal parameter point.

Chapter 15

P292 — Independent reservoir replication

The primary endpoint was NARMA10 prediction. Twenty-four independent input realizations were evaluated against twelve matched controls per realization: six isospectral controls and six spectral-radius controls.

quantity	result
wins against every control	$13/24 = 0.5417$
median control percentile	1.000
median advantage over best control	0.000962
fifth percentile of the advantage	-0.00713
median delay-task percentile	0.4167
median nonlinear-task percentile	0.4167
median parity-task percentile	0.0833

During falsification, the original NARMA input convention overflowed on some replications and generated non-finite targets. That implementation was rejected. P292 was rerun with the standard stable independent NARMA driving range $u_t \in [0, 0.40]$; the reservoir-control inputs otherwise remain matched. The correction is methodological, not cosmetic: non-finite trials cannot count as supporting evidence.

Verdict. [\[Strong evidence\]](#) A small primary-task advantage replicates only partially. The negative lower-tail advantage and weak secondary-task percentiles refute any claim of universal computational superiority.

Chapter 16

P293 — Resource-explicit collision-model erasure

P293 does not derive temperature, energy, time, or a work scale from the dimensionless strict generator. It supplies them explicitly:

$$k_B = 1, \quad T = 2, \quad \text{maximum gap} = 24,$$

together with 800 fresh ideal Gibbs ancillas and a classical work meter. A finite sequence of quenches and thermalizing collisions gives:

ledger item	value
final excited-state probability	6.144×10^{-6}
erased information	0.693067 nat
work supplied	1.393644
heat dumped	1.393644
generalized Landauer lower bound	1.386135
work above bound	0.007509
entropy production	0.003755
first-law residual	4.22×10^{-15}

The protocol demonstrates how informational erasure becomes thermodynamic only after a bath state, Hamiltonian gaps, temperature, interaction schedule, and work convention have been added.

Theorem (conditional finite resource ledger). For the supplied collision-model protocol, work, heat, information loss, and entropy production obey the finite first- and second-law ledger to numerical precision.

Boundary. This does not identify $k_B T$, a Hamiltonian scale, or a physical clock from FIN. It is a valid information-to-physics bridge with all conversion resources exposed.

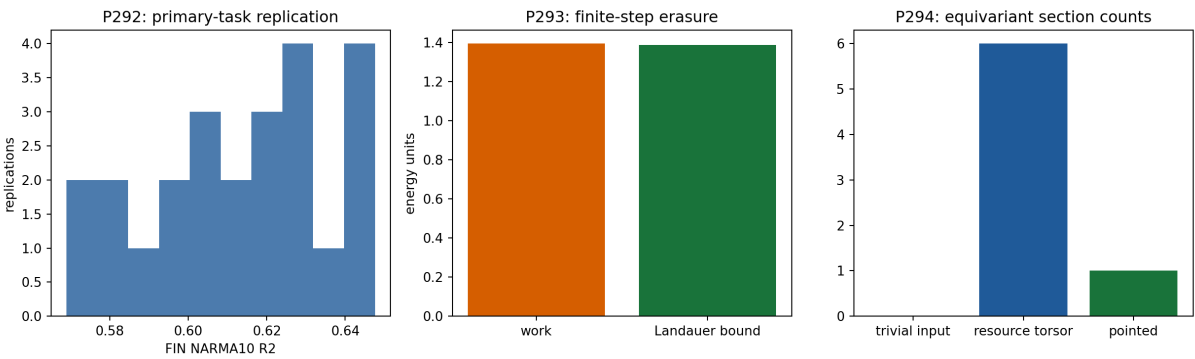


Figure 16.1: Reservoir replication, the thermodynamic resource ledger, and pointed-torsor counts.

Chapter 17

P294 — The pointed physicalization resource

Let

$$G = \mathbb{Z}_3 \times \mathbb{Z}_2$$

act regularly on the torsor $T = G$. A section from a trivial one-point G -space would have to choose a fixed point of T , but a nontrivial regular torsor has no fixed point. The finite enumeration gives:

$$\# \text{Sec}_G(*, T) = 0, \quad \# \text{End}_G(T) = 6.$$

After adjoining a distinguished point $t_0 \in T$, the pointed equivariant endomorphism is unique:

$$\# \text{End}_{G,*}(T, t_0) = 1.$$

This motivates the newly constructed object

$$\boxed{\mathfrak{P} = (T, t_0)}$$

called the **Pointed Physicalization Resource**. It is the minimal operational datum that turns a symmetry-related family of equivalent conventions into a unique convention-compatible map.

Theorem. [Proven] Pointing the regular finite torsor removes the translation ambiguity in its pointed equivariant self-map.

Impossibility boundary. [Proven] Equivariance alone cannot derive the point. Thus $\mathfrak{P}$ can encode a prepared origin, clock zero, detector label, chirality choice, or calibration reference, but it cannot be used to claim that strict FIN internally generates any of them.

Chapter 18

Theoretical objects constructed or sharpened

Object	Definition or role	Status	What it does not supply
O40 Compactly Regularized Stieltjes Inverse	bounded positive pole/residue inverse problem	[Proven] existence; [Strong evidence] numerics	existence or stable residues
O41 Lean Rational Kernel Certificate	Schur exact 3×3 rational nested/direct Schur witness	[Proven]	general matrix theorem
O42 Loss-Augmented Naimark Current Readout	Naimark dilation followed by a stochastic detector channel	[Proven] bra; [Strong evidence] counts	algebraic apparatus construction
O43 Isotropic Complement RG Functor	$B \mapsto EBE^* + \bar{\lambda}(I - EE^*)$	[Proven] PSD/rank facts	strict fixed-point closure
O44 Two-Step Process Ledger	Quantum multitime decomposition of relative entropy loss	[Proven], conditional	con-arbitrary process-tensor tomography
O45 Legacy Fractal Compression Atom	$d \mapsto d^{4 \ln 2}$ in the legacy denominator	[Refuted] sufficient	as generic bridge obstruction
O46 External Evidence Admission Gate	manifest/schema/hash/hold-out preconditions for P242	[Proven]	as physical data
O47 Detector-Likelihood Flux Design	apparatus-aware optimization of event likelihood	[Strong evidence]	evi-intrinsic FIN flux
O48 Dirichlet-Null Fingerprint Risk	ensemble-conditioned false-positive probability	[Strong evidence]	evi-distribution-free uniqueness
O49 Interventional Time Mechanism Atlas	Three-finite diagnostic map for four mechanism classes	[Strong evidence]	evi-arbitrary causal identification
O50 Maximin Control Design	Adaptive maximize the weakest local information direction	[Strong evidence]	evi-global input optimality
O51 Primary-Task Replication Ledger	paired seeds and matched control families	[Strong evidence]	evi-universal computational advantage
O52 Collision-Model Erasure Resource Ledger	work/heat/information accounting with exposed resources	[Proven], conditional	con-emergent dimensional constants
O53 Pointed Physicalization Resource	a torsor plus a supplied distinguished point	[Proven] theorem	finite derivation of the point

The strongest new conceptual result is O53. It makes the mathematics-to-physics boundary precise: a symmetric mathematical object can support many equivalent operational coordinatizations, while a physical realization necessarily contains a preparation/calibration event that points one of them. The pointing may be supplied by an experiment, boundary condition, environment, or new symmetry-breaking law. It is not produced by the spectral theorem.

Chapter 19

Falsification register

The following attractive conclusions did not survive:

1. **"Compact Stieltjes priors solve spectral inversion."** Refuted. They give existence and prevent runaway solutions, but pole/residue recovery remains ill-conditioned.
2. **"Rank-complete lifted RG is the strict kernel."** Refuted by nonzero strict-fingerprint residuals.
3. **"Fractal compression alone completes legacy into strict."** Refuted for the one canonical atom tested by negative mean-zero and hidden spectra.
4. **"No direct null hits means the false-positive rate is zero."** Refuted as an inference; importance sampling resolves a small but nonzero ensemble-conditioned probability.
5. **"Three-time observations identify the true mechanism."** Refuted beyond the frozen panel; the result is finite-class discrimination.
6. **"Random or broadband intervention is necessarily best."** Refuted in the frozen local design; a balanced two-level pulse wins.
7. **"The FIN reservoir is universally superior."** Refuted by incomplete primary-task wins and weak secondary-task ranks.
8. **"Landauer converts dimensionless information into physics without extra structure."** Refuted. The successful ledger explicitly supplies temperature, gaps, bath, work meter, and protocol.
9. **"Equivariance selects an operational origin."** Refuted by the absence of a section from a trivial input to a nontrivial regular torsor.
10. **"A pointed torsor derives the selector."** Refuted as a logical direction. Pointing removes ambiguity only after the point is supplied.

Chapter 20

Present mathematical and physical state of FIN

The repository's most defensible core remains finite spectral operator theory with adaptive, informational, and operational extensions. The present chain is

$$\text{strict kernel} \Rightarrow A \Rightarrow \{e^{-itA}, e^{-tA}, (A + zI)^{-1}, E_A\}$$

followed, only with extra data, by

$$(A, \mathcal{C}) \xrightarrow{\mathfrak{D}} \text{prepared and measured process} \xrightarrow{\mathfrak{C}, \mathfrak{S}, \mathfrak{P}} \text{dimensional, sector-resolved physical model.}$$

Here:

- $\mathfrak{D}$ contains state preparation, clock protocol, instruments, environment, detector response, and immutable event records;
- $\mathfrak{C}$ contains dimensional conversion resources such as energy and time scales;
- $\mathfrak{S}$ specifies the physical sector and observable semantics;
- $\mathfrak{P}$ supplies a point in the relevant calibration or symmetry torsor.

P281–P294 do not close QW-2191. They do not derive a canonical UV/length unit, the strict damping source, a role-safe legacy amplitude, a full nonproxy Lagrangian/EOM, Standard Model fields, general relativity, L_{total} , or a Theory of Everything.

The round does, however, replace several vague gaps by executable statements:

- the inverse spectral obstruction is conditioning, not absence of a finite minimizer;
- an operational current measurement has an explicit dilation and detector channel;
- resource-explicit thermodynamics is possible but conditional;
- physical convention selection is naturally represented by a pointed torsor;
- the point itself is the irreducible missing datum unless a new internal symmetry-breaking theorem is found.

Chapter 21

Recommended Programs P295–P308

The following programs are new bounded moves, not replays of closed FAR lanes. Probabilities are estimates of producing a scientifically decisive result, not probabilities that FIN is physically correct.

Rank	Program	Proposed study	Decisive output	Success probability
1	P295	Multi-probe operator-valued Stieltjes recovery with minimax bounds	identifiability/stability theorem or matched lower bound	0.80
2	P296	General positive Schur-complement theorem in Lean/Mathlib	reusable machine-checked theorem extending O41	0.78
3	P297	Compile O42 into an optical/unitary mesh with tolerance propagation	component-level executable apparatus specification	0.72
4	P304	Adversarial out-of-family mechanism identifiability	equivalence classes and a no-go/witness criterion	0.72
5	P303	Coarea/Laplace asymptotics for the Dirichlet fingerprint rare event	analytic exponent/prefactor checked against P289	0.70
6	P299	General process-tensor ledger and finite Markov-order tomography	theorem plus reconstruction protocol	0.68
7	P302	Sequential detector tomography with calibration uncertainty	joint design for h and nuisance parameters	0.67
8	P298	Uniqueness/no-go classification of complement RG completions	classification of admissible target-independent complements	0.65
9	P305	Nonlinear adaptive model discovery under interventions	out-of-panel recovery and falsification suite	0.63
10	P307	Embed $A_{\text{phys}} = \kappa A$ into collision thermodynamics	identify scale-free predictions and scale-dependent observables	0.62
11	P308	General pointed-torsor nonsection/minimal-resource theorem	abstract theorem and Lean formalization	0.60
12	P300	Phase-fixed denominator-only legacy completion no-go class	proof or counterexample for a defined bridge class	0.58
13	P306	External/hardware reservoir replication	preregistered effect estimate against matched controls	0.45
14	P301	Admit and execute a real P241 blind-custody laboratory bundle	P242 accept/reject report without model repair	0.35

21.1 Stop rules

- P295 stops with a minimax lower bound if residue stability cannot be made uniform.
- P297 stops if the compiled mesh cannot preserve POVM completeness within a declared calibration budget.
- P298 may not target-code the strict answer into the complement.
- P300 studies one explicitly stated completion class and may not imply a generic legacy-to-strict no-go.
- P301 remains blocked until an external provider supplies the production manifest, event rows, hashes, custody roles, and frozen hold-out.

- P306 must freeze the primary endpoint and matched controls before hardware data are unblinded.
- P307 must expose κ ; it may not rename an inserted dimensional scale as an emergent prediction.
- P308 may prove that pointing is minimal, but may not claim that strict FIN derives the point.

Chapter 22

Recommended execution order

The immediate mathematical sequence is:

$$\boxed{P295 \rightarrow P296 \rightarrow P303 \rightarrow P304 \rightarrow P298}.$$

The operational sequence is:

$$\boxed{P297 \rightarrow P302 \rightarrow P301/P306 \text{ when external data exist}}.$$

The information-to-physics sequence is:

$$\boxed{P299 \rightarrow P307 \rightarrow P308}.$$

P300 is a bounded bridge-class falsification and should not displace these higher-yield programs.

Chapter 23

Reproducibility

The release contains:

- `fin_programs_281_294.py` — deterministic research runner;
- `test_fin_programs_281_294.py` — 19 regression and guardrail tests;
- `FIN_Programs_281_294_Schur_Core.lean` — dependency-free Lean 4.28 certificate;
- `FIN_Programs_281_294_Results.json` — full machine-readable results;
- `FIN_Programs_281_294_Summary.csv` — one-row-per-program summary;
- five detailed CSV audit tables;
- five generated figures;
- this English report and its PDF rendering.

The numerical suite is reproduced by setting `MPLCONFIGDIR` to `/tmp/matplotlib-p281-294` and running `python3 -m unittest -v test_fin_programs_281_294.py`.

The Lean certificate prints the nested and direct rational Schur values $13/5$ and exits successfully. Random generators use frozen seeds. P292 uses a separate stable NARMA input stream after the overflow falsification described in Section 15.

Chapter 24

Final conclusion

P281–P294 advance FIN most strongly by clarifying boundaries.

The spectral generator supports exact wave, diffusion, resolvent, measurement, reduction, and information-ledger constructions. These are consequences of a finite positive operator plus supplied contexts. They are not by themselves a physical law.

The smallest new bridge object found in this round is not a new kernel. It is the Pointed Physicalization Resource

$$\mathfrak{P} = (T, t_0).$$

It expresses a general fact: experimental physics needs a prepared reference inside a symmetry family. Once supplied, that reference can make an operational map unique. Mathematics based only on the unpointed symmetric object cannot select it.

Accordingly, the deepest conclusion surviving falsification is:

FIN currently defines a coherent finite spectral-information dynamics,
but physicalization requires explicit resources
 $\mathfrak{D}, \mathfrak{C}, \mathfrak{S}, \mathfrak{P}.$

This is a constructive boundary, not a dismissal. It tells the next research round exactly what must be proved, calibrated, or externally supplied—and what must no longer be inferred from spectral coherence alone.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/matplotlib-p281-294 \  
python3 -m unittest -v test_fin_programs_281_294.py
```

Expected result: 19 tests passed, 0 failed. The suite recompiles the dependency-free Lean 4.28 rational Schur witness. Synthetic and conditional studies are not laboratory evidence. P287 certifies that no independent production P241 event bundle was available and therefore P242 was not authorized.

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